

Dasari Naga Raju

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Research Interests

Computational pathology with a focus on pathology foundation models, weakly supervised learning, and biologically grounded representation learning for molecular biomarker and phenotype prediction from H&E whole-slide images. I am currently focused on developing robust and transferable methods for spatial tissue modelling, cross-site and cross-cancer generalization, domain adaptation, and interpretable computational pathology across diverse cancer cohorts through **Project TARAA**.

Education

International Institute of Information Technology Bangalore M.S. by Research in Information Technology <i>Research Area: Computational Pathology 59 of 75 credits in supervised thesis research</i>	Sept 2021 – July 2024 CGPA: 3.16 / 4.0
Jawaharlal Nehru Technological University Gurajada, Vizianagaram B.Tech. in Electronics and Communication Engineering	July 2016 – Sept 2020 CGPA: 7.6 / 10

M.S. Thesis

DeepNucleiNet: Encoding Spatial Nuclear Patterns for Data-Efficient Necrosis Detection in Computational Neuropathology *Published at MICCAI 2026 Workshop on BrainWorks [Paper]*

- Demonstrated that explicitly encoding nuclear spatial patterns provides discriminative information beyond tissue appearance and general-purpose pathology foundation models in data-limited settings; DeepNucleiNet achieved a mean **F1-score of 0.942**, compared with **0.884** for UNI2-h and **0.730** for Virchow2 under strict inter-patient evaluation.
- Showed that nuclear depletion is a transferable morphological signal across related CNS tumours: the meningioma-trained model generalized without retraining to TCGA-GBM with an **F1-score of 0.943** and produced no necrosis predictions across **85,632** TCGA-LGG tiles, supporting cross-tumour generalization beyond the training cohort.

Publications

1. **Dasari Naga Raju**, Tripti Bameta, “State-Aware Interaction MIL for Rare Joint Molecular Phenotype Prediction in Colorectal Cancer and Lung Adenocarcinoma.” *Accepted at NeurIPS 2026 Workshop on AISC: Cancer Pathology Foundation Models.* [Paper] [Code]
2. **Dasari Naga Raju**, “Conserved Immune Topology Improves Pathology Foundation Model Generalization for Cross-Cancer MSI-H Prediction.” *Published at ECCV 2026 Workshop on Medical Foundation Models and Benchmarks (MedFMB).* [Paper] [Code]
3. **Dasari Naga Raju**, “What Do Pathology Foundation Models Learn Beyond Gleason Grade in Prostate Cancer?” *Published at MICCAI 2026 Workshop on Computational Pathology and Multimodal Data (COMPAYL).* [Paper] [Code]
4. **Dasari Naga Raju**, T.K. Srikanth, Shilpa Rao, Ramesh Kestur, Anita Mahadevan, “DeepNucleiNet: Encoding Spatial Nuclear Patterns for Data-Efficient Necrosis Detection in Computational Neuropathology.” *Published at MICCAI 2026 Workshop on BrainWorks.* [Paper] [Code]
5. V.K. Kancharla, **Dasari Naga Raju**, P. Kaveti, “Self-Supervised Denoising Reconstruction Outperforms Foundation Models for Capsule Endoscopy Abnormality Classification.” *Published at MICCAI 2026 Workshop on AI for Safe Surgery (SafeSurg).* [Paper]
6. **Dasari Naga Raju**, V.K. Kancharla, P. Kaveti, “Surgical Foundation Model Ensembles for Predicting Annotator Disagreement in Critical View of Safety Assessments.” *Published at MICCAI 2026 Workshop on Clinical-driven Robotics and Embodied AI Technology (CREATE).* [Paper]
7. V.K. Kancharla, **Dasari Naga Raju**, P. Kaveti, “Weave Before You Weigh: Interleaved Head Attention for Multimodal Physiological Stress Recognition in the Wild.” *Published at ECCV 2026 11th Workshop and Competition on Affective & Behavior Analysis in-the-Wild.*

Research Experience

Project Scientist (Contract), Medical College of Wisconsin *Feb 2025 – Present*
Research Supervisor: Dr. Pradeep Javarayee, MD, Epileptologist, Children’s Wisconsin & Medical College of Wisconsin

- Developing deep learning models for seizure detection across multi-site clinical data, with a focus on robust

generalization across hospitals and heterogeneous acquisition settings.

- Designing and evaluating cross-site validation strategies to identify failure modes, quantify domain shift, and improve model reliability across independent clinical cohorts.

Research Associate, MINRO Lab, IIIT Bangalore

Aug 2024 – Jan 2025

- Extended DeepNucleiNet to independent WSI cohorts spanning Meningioma, TCGA-GBM, NIMHANS GBM, TCGA-LGG, DeepHisto, and TiGER, demonstrating cross-cohort robustness without retraining; achieved F1-scores of **0.943** on GBM, **0.843** on DeepHisto, and **0.635** on TiGER.
- Evaluated grading-aware behaviour on Low-Grade Glioma as a controlled negative cohort, producing no necrosis predictions across **85,000+** tiles and demonstrating consistency with the expected absence of necrosis in lower-grade disease.

Clinical Data Collaborator, NIMHANS Hospital × IIIT Bangalore

Sept 2021 – Aug 2022

- Conducted a year-long on-site collaboration with the Department of Neuropathology at NIMHANS for WSI data management, gigapixel image processing, and expert annotation of Meningioma, Glioblastoma, and Low-Grade Glioma specimens.
- Worked closely with neuropathologists to annotate necrotic regions, cellular populations, white space, and tissue regions of interest using **ImageScope**; the resulting pathology insights directly informed the biologically grounded feature design of DeepNucleiNet.

Research Associate, Mirai Research Innovation Institute, Japan

Oct 2020 – Aug 2021

- Developed a CNN-based visual inspection pipeline using ResNet50 and transfer learning for detecting missing electronic components in PCB imagery, improving classification accuracy from **82% to 94%**.

Technical Skills

Digital Pathology	OpenSlide, QuPath, ImageScope, HALO, StarDist, whole-slide image processing
Spatial & Tissue Analysis	Cell phenotyping, tissue-level spatial feature extraction, biologically grounded spatial priors, tumour microenvironment characterisation
Deep Learning & MIL	PyTorch, TensorFlow/Keras, scikit-learn, Vision Transformers, multiple-instance learning, UNI2-h, Virchow2
Image Representation	Tissue segmentation, object detection, representation learning, domain adaptation, Grad-CAM, t-SNE
Biomedical Signals	MNE-Python, EEG preprocessing and analysis